

Sleep, Academic Pressure and Depression Among Students

Sapir Hai, Lea Sayada
Statistical Theory Project
Bar-Ilan University

Abstract—Depression among university students has become an increasing public health concern, with academic pressure and sleep habits recognized as two major factors affecting students' mental health. This study investigated the relationship between sleep duration, academic pressure, and depression using the Student Depression Dataset, which contains information from more than 140,000 students. Statistical analyses included a Chi-square test, a Mann–Whitney U test, logistic regression, and confidence intervals.

The results revealed a significant association between sleep duration and depression ($\chi^2 = 276.85$, $df = 4$, $p < 0.001$). In addition, students with depression reported significantly higher academic pressure than students without depression (Mann–Whitney $U = 145,556,846.5$, $p < 0.001$). Logistic regression further showed that academic pressure was a strong predictor of depression ($OR = 2.21$, $p < 0.001$), while sleeping more than eight hours was associated with lower odds of depression ($OR = 0.69$, $p = 0.001$). However, the interaction between sleep duration and academic pressure was not statistically significant.

Overall, our findings suggest that both sleep duration and academic pressure are independently associated with depression among university students. These results emphasize the importance of promoting healthy sleep habits and reducing academic stress to improve students' mental well-being.

I. INTRODUCTION

Depression is one of the most common mental health disorders among university students and has become an increasing public health concern worldwide [1]. Academic life exposes students to numerous stressors, including demanding coursework, examinations, financial concerns, and uncertainty about future careers. These factors may negatively affect students' psychological well-being and increase their risk of developing depressive symptoms.

Among the many factors associated with depression, sleep duration has received considerable attention. Adequate sleep is essential for cognitive performance, emotional regulation, and physical health. Previous studies have shown that insufficient or poor-quality sleep is associated with higher levels of stress, anxiety, and depression among university students [2]. At the same time, academic pressure has consistently been identified as one of the strongest predictors of psychological distress and depressive symptoms among university students [3].

Although both sleep duration and academic pressure are known to influence mental health, their combined effect has been less extensively investigated. In particular, it remains unclear whether sleep duration moderates the relationship between academic pressure and depression. Understanding this interaction may help identify protective factors and contribute

to the development of interventions aimed at improving students' mental well-being.

In this study, we analyzed the Student Depression Dataset, containing information from more than 140,000 students. The dataset includes demographic, academic, and lifestyle variables such as age, gender, cumulative grade point average (CGPA), sleep duration, academic pressure, and depression status.

The primary research question of this study was:

Does sleep duration moderate the relationship between academic pressure and depression among students?

To answer this question, we examined the association between sleep duration and depression using a Chi-square test, compared academic pressure between depressed and non-depressed students using a Mann–Whitney U test, and evaluated the combined effects of academic pressure, sleep duration, and their interaction through logistic regression. Confidence intervals were also calculated to assess the precision of the estimated effects. Together, these analyses provide a comprehensive statistical approach for investigating the relationship between sleep habits, academic pressure, and depression among university students.

II. RESULTS

A. Dataset Overview

The analysis was conducted using the Student Depression Dataset. After data preprocessing and cleaning, observations with missing or invalid values were excluded, resulting in a final analytical sample of 27,901 students.

The dataset used for the analyses included the variables relevant to the research question, namely sleep duration, academic pressure, and depression status.

B. Association Between Sleep Duration and Depression

Our first hypothesis was that sleep duration is associated with depression among university students.

To test this hypothesis, a Chi-square test of independence was performed. The null hypothesis stated that sleep duration and depression status are independent, whereas the alternative hypothesis stated that an association exists between these variables.

The results revealed a statistically significant association between sleep duration and depression ($\chi^2 = 276.85$, $df = 4$, $p < 0.001$). Therefore, the null hypothesis was rejected,

indicating that depression rates differ significantly across sleep duration categories.

As shown in Figure 1, the depression rate varied across sleep-duration groups, with the highest rate observed among students sleeping less than five hours.

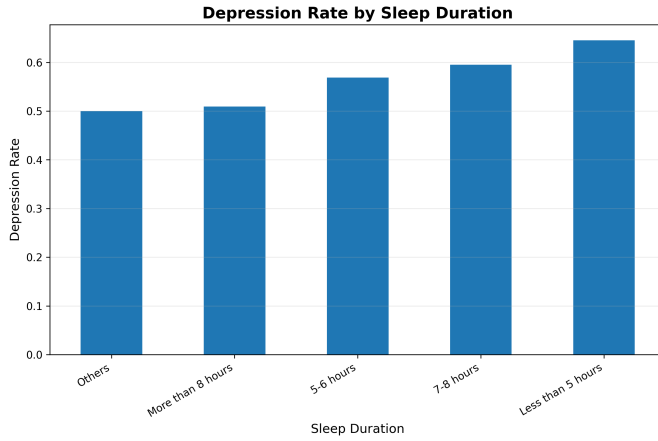


Fig. 1. Depression rate according to sleep duration. Depression rates differ across sleep-duration categories, with the highest rate observed among students sleeping less than five hours.

C. Academic Pressure Among Students With and Without Depression

Our second hypothesis was that students with depression experience higher academic pressure than students without depression.

Since academic pressure is an ordinal variable, a Mann–Whitney U test was used to compare the two independent groups.

The analysis revealed a statistically significant difference in academic pressure between students with and without depression ($U = 145,556,846.5$, $p < 0.001$). The median academic pressure was 4 among students with depression and 2 among students without depression.

These results indicate that students with depression tend to report substantially higher levels of academic pressure. This difference is also visible in Figure 2.

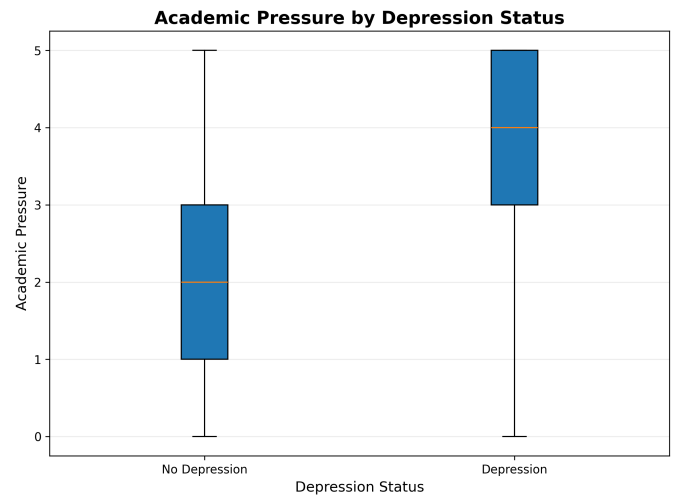


Fig. 2. Distribution of academic pressure according to depression status. Students with depression reported higher academic pressure than students without depression.

D. Logistic Regression Analysis

A logistic regression model was fitted to evaluate the combined effects of academic pressure, sleep duration, and their interaction on the probability of depression.

The results indicated that academic pressure was a statistically significant predictor of depression ($\beta = 0.791$, $OR = 2.21$, $p < 0.001$). This indicates that, for each one-unit increase in academic pressure, the odds of depression increased by approximately 2.21 times while holding sleep duration constant.

Students who reported sleeping more than eight hours had significantly lower odds of depression ($OR = 0.69$, $p = 0.001$). The remaining sleep-duration categories were not statistically significant.

To address the primary research question, interaction terms between academic pressure and sleep duration were included in the logistic regression model. None of the interaction terms were statistically significant ($p > 0.05$), indicating that sleep duration does not significantly moderate the relationship between academic pressure and depression.

Figure 3 illustrates this relationship. Although depression rates increased with academic pressure across all sleep-duration groups, the trends were very similar. This visual pattern is consistent with the logistic regression results, which showed no statistically significant interaction effect.

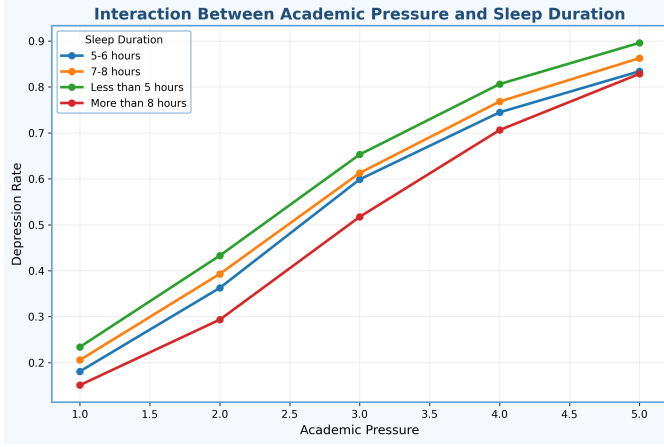


Fig. 3. Interaction between academic pressure and sleep duration. Depression rates increased with academic pressure across all sleep-duration groups. The similar trends across groups are consistent with the non-significant interaction terms observed in the logistic regression model.

Table I summarizes the odds ratios (OR), 95% confidence intervals, and p-values obtained from the logistic regression model.

TABLE I
LOGISTIC REGRESSION RESULTS

Variable	OR	95% CI	p-value
Academic Pressure	2.21	2.10–2.31	< 0.001
Sleep: 7–8 hours	1.09	0.89–1.33	0.416
Sleep: Less than 5 hours	1.19	0.98–1.45	0.082
Sleep: More than 8 hours	0.69	0.55–0.85	0.001
Interaction terms	—	—	> 0.05

Overall, the logistic regression analysis indicates that both academic pressure and sleep duration were independently associated with depression. However, the interaction analysis did not provide evidence that sleep duration significantly modifies the relationship between academic pressure and depression.

III. METHODS

A. Dataset

The analyses were performed using the Student Depression Dataset obtained from Kaggle. The original dataset contains demographic, academic, and lifestyle information for more than 140,000 students, including age, gender, cumulative grade point average (CGPA), sleep duration, academic pressure, and depression status.

The data were imported and processed using Python. Data preprocessing included selecting the variables relevant to the research question, removing observations with missing or invalid values, and formatting categorical variables for statistical analysis.

After preprocessing, the final analytical sample consisted of 27,901 students, which was used in all subsequent statistical analyses.

B. Software and Libraries

All analyses were performed in Python using Jupyter Notebook. The Pandas library was used for data manipulation and preprocessing, NumPy for numerical computations, SciPy for statistical hypothesis testing, Statsmodels for logistic regression, and Matplotlib for data visualization.

C. Chi-Square Test

A Chi-square test of independence was used to examine the association between sleep duration and depression status.

The observed frequencies were compared with the expected frequencies under the assumption of independence. The test statistic, degrees of freedom, and p-value were calculated to determine whether a significant association existed between the two categorical variables.

D. Mann–Whitney U Test

Academic pressure is measured on an ordinal scale; therefore, the Mann–Whitney U test was used to compare academic pressure between students with and without depression.

This non-parametric test compares the distributions of two independent groups without assuming normality. The U statistic and corresponding p-value were computed to evaluate whether the distributions differed significantly.

E. Confidence Intervals

A bootstrap resampling procedure was used to estimate 95% confidence intervals for the median difference in academic pressure between students with and without depression.

Bootstrap samples were repeatedly generated with replacement from the original data, allowing estimation of the sampling distribution without relying on parametric assumptions.

F. Logistic Regression

A logistic regression model was fitted using the Statsmodels library to estimate the probability of depression.

Depression status was defined as the binary response variable, while academic pressure and sleep duration were included as explanatory variables. An interaction term between academic pressure and sleep duration was added to evaluate whether sleep duration modifies the relationship between academic pressure and depression.

Model coefficients, odds ratios (OR), 95% confidence intervals, and p-values were calculated to assess the contribution of each predictor.

IV. DISCUSSION

The primary objective of this study was to examine the relationship between sleep duration, academic pressure, and depression among university students. Using several statistical methods, we found that both sleep duration and academic pressure were significantly associated with depression.

The Chi-square test revealed a significant association between sleep duration and depression, indicating that depression rates differed across sleep-duration categories. In addition, the Mann–Whitney U test showed that students with depression

reported significantly higher levels of academic pressure than students without depression. Logistic regression further confirmed that academic pressure was a strong predictor of depression. However, the interaction between sleep duration and academic pressure was not statistically significant, suggesting that sleep duration does not significantly modify the effect of academic pressure on depression.

These findings suggest that sleep duration and academic pressure are independently associated with students' mental health. Although adequate sleep remains important for overall well-being, our results indicate that academic pressure may have a stronger association with depression than sleep duration in the present dataset.

This study has several limitations. First, the dataset is observational; therefore, causal relationships cannot be established. Second, several variables, including sleep duration and academic pressure, were self-reported, which may introduce reporting bias. In addition, other factors that may influence depression, such as socioeconomic status, family environment, personality traits, and previous mental health conditions, were not included in the analysis.

Future research could investigate additional variables associated with depression, analyze longitudinal data to better understand causal relationships, and evaluate whether interventions aimed at improving sleep habits or reducing academic pressure lead to measurable improvements in students' mental health.

Overall, this study demonstrates the usefulness of statistical methods in identifying factors associated with depression among university students. The findings highlight the importance of considering both lifestyle and academic factors when developing strategies to improve students' psychological well-being.

REFERENCES

- [1] Y. Liu, J. Chen, K. Chen, J. Liu, and W. Wang, "The associations between academic stress and depression among college students: A moderated chain mediation model of negative affect, sleep quality, and social support," *Acta Psychologica*, vol. 239, p. 104014, 2023.
- [2] J. Dinis and M. Bragança, "Quality of sleep and depression in college students: A systematic review," *Sleep Science*, vol. 11, no. 4, pp. 290–301, 2018.
- [3] A. A. Alhamed, "The link among academic stress, sleep disturbances, depressive symptoms, academic performance, and the moderating role of resourcefulness in health professions students during covid-19 pandemic," *Journal of Professional Nursing*, vol. 46, pp. 83–91, 2023.